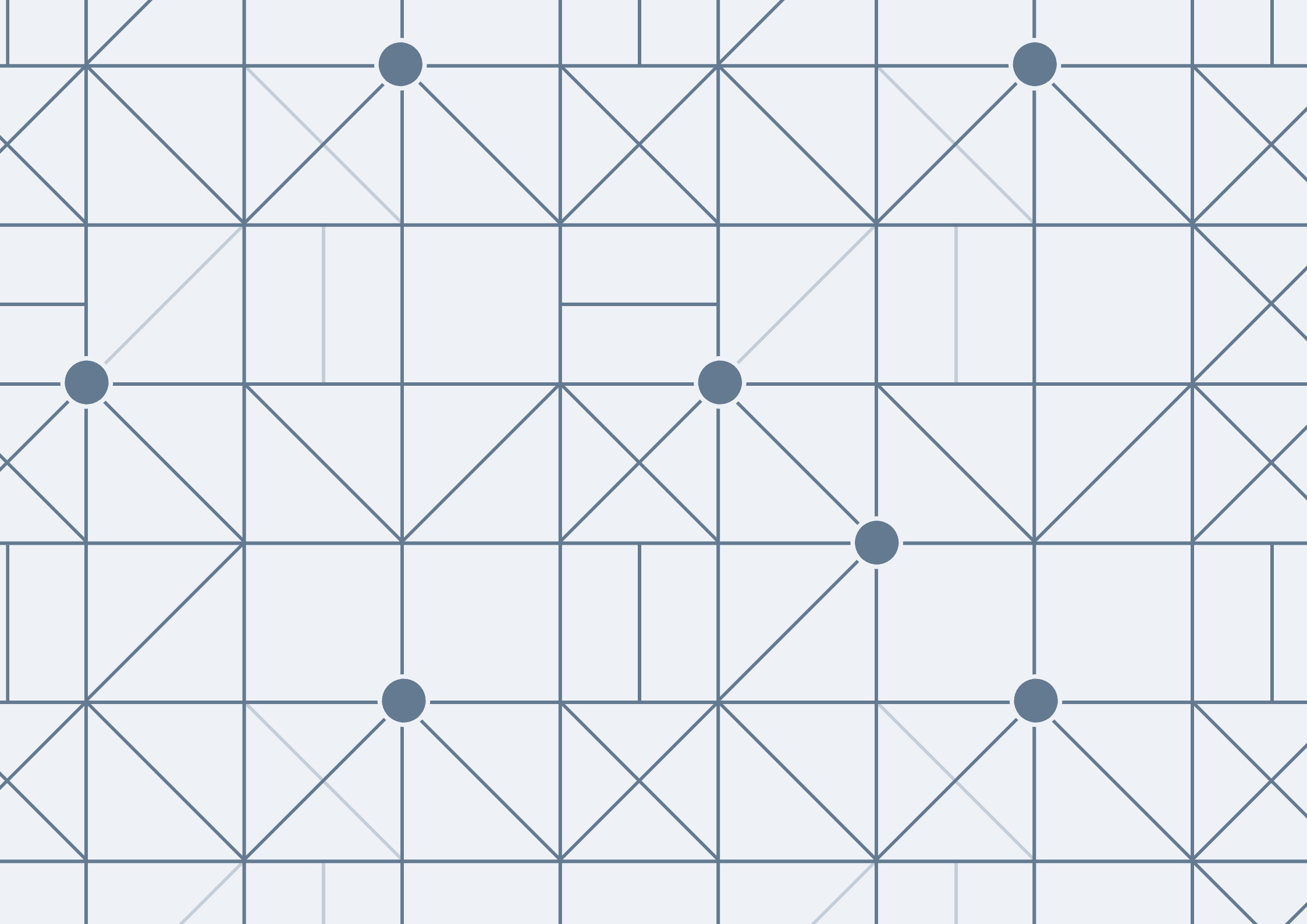




test

Derive -

$$f(n) = \sum_{i=1}^n i^2 = 1^2 + 2^2 + \dots + n^2$$
$$= \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^n i^3 = S$$

$$\sum_{i=1}^n (i+1)^3 = S + (n+1)^3 - 1^3$$

$$\cancel{S} + 3 \sum i^2 + \underbrace{3 \sum i}_{\text{cancel}} + \underbrace{\sum 1}_{\text{cancel}} = \cancel{S} + (n+1)^3 - 1$$

$$3 \sum i^2 = (n+1)^3 - 1 - n - 3 \frac{n(n+1)}{2}$$

$$\sum i^2 = \frac{(n+1) \left[(n+1)^2 - 1 - \frac{3n}{2} \right]}{3}$$

$$f(i, w_0, w_1) = 1 - \left(1 - \left(\frac{y_i - (w_0 + w_1 x_i)}{100} \right)^2 \right)^3$$

$$\text{Let } e_i = y_i - (w_0 + w_1 x_i)$$

$$f(i) = 1 - \left(1 - \left(\frac{e_i}{100} \right)^2 \right)^3$$

$$\frac{\partial f}{\partial e_i} = -3 \left(1 - \left(\frac{e_i}{100} \right)^2 \right)^2 \left[-2 \frac{e_i}{100} \right]$$

$$= \frac{6}{100} \left[1 - \left(\frac{e_i}{100} \right)^2 \right]^2 e_i$$

$$\frac{\partial f}{\partial w_0} = \frac{6}{100} \left[1 - \left(\frac{e_i}{100} \right)^2 \right]^2 (y_i - (w_0 + w_1 x_i)) (-1)$$

$\frac{\partial f}{\partial w_1} = \frac{6}{100} \left[1 - \left(\frac{e_i}{100} \right)^2 \right]^2 (y_i - (w_0 + w_1 x_i)) (-x_i)$

$$Rsq(\vec{w}) = \frac{1}{n} \|\vec{y} - X\vec{w}\|^2$$

$\vec{y}, X$ come from data

$$X = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} = \begin{bmatrix} & \\ \vdots & \vdots \\ & \end{bmatrix}$$

departure,
departure₂
⋮
departure_n

$$\vec{y} = \begin{bmatrix} \text{commute}_1 \\ \text{commute}_2 \\ \vdots \\ \text{commute}_n \end{bmatrix}$$

$$\begin{aligned}
 X\vec{w} &= \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix} \begin{matrix} \text{departure}_1 \\ \text{departure}_2 \\ \vdots \\ \text{departures}_n \end{matrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} \\
 &= w_0 \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + w_1 \begin{bmatrix} \text{dept}_1 \\ \text{dept}_2 \\ \vdots \\ \text{dept}_n \end{bmatrix}
 \end{aligned}$$

